

Integrals: Formula Sheet & Reference

Class 12 Mathematics – Chapter 7 Complete Reference & Formulas

1 Indefinite Integrals: Standard Formulas & Algebraic Properties

1. Standard Algebraic, Exponential & Trigonometric Integrals:

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), & \int dx &= x + C \\ \int \frac{1}{x} dx &= \ln |x| + C, & \int e^x dx &= e^x + C, & \int a^x dx &= \frac{a^x}{\ln a} + C \\ \int \cos x dx &= \sin x + C, & \int \sin x dx &= -\cos x + C \\ \int \sec^2 x dx &= \tan x + C, & \int \operatorname{cosec}^2 x dx &= -\cot x + C \\ \int \sec x \tan x dx &= \sec x + C, & \int \operatorname{cosec} x \cot x dx &= -\operatorname{cosec} x + C \\ \int \frac{dx}{\sqrt{1-x^2}} &= \sin^{-1} x + C = -\cos^{-1} x + C_1 \\ \int \frac{dx}{1+x^2} &= \tan^{-1} x + C = -\cot^{-1} x + C_1\end{aligned}$$

2. Fundamental Algebraic Properties of Indefinite Integrals:

- **Inversion Relation:** $\frac{d}{dx} \left[\int f(x) dx \right] = f(x)$ and $\int f'(x) dx = f(x) + C$.
- **Linearity Property:** $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$.
- **Scalar Multiplication:** $\int k \cdot f(x) dx = k \int f(x) dx \quad (k \in \mathbb{R})$.

2 Integration by Substitution & Four Trigonometric Forms

1. Variable Transformation Formula:

$$\int f(x) dx = \int f(g(t)) g'(t) dt \quad (\text{putting } x = g(t), dx = g'(t) dt)$$

2. Four Standard Integrals Derived via Substitution:

$$\int \tan x dx = \ln |\sec x| + C = -\ln |\cos x| + C$$

$$\int \cot x dx = \ln |\sin x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x - \cot x| + C = \ln \left| \tan \left(\frac{x}{2} \right) \right| + C$$

3 Integrals of Particular Algebraic Functions

$$1. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C$$

$$2. \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right| + C$$

$$3. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$4. \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$5. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

$$6. \int \frac{dx}{\sqrt{x^2 + a^2}} = \ln \left| x + \sqrt{x^2 + a^2} \right| + C$$

4 Quadratic Integrals & Linear / Quadratic Forms

1. Standard Quadratic Forms:

$$\int \frac{dx}{ax^2 + bx + c} \quad \text{or} \quad \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

Method: Complete the square in the denominator: $ax^2 + bx + c = a \left[\left(x + \frac{b}{2a} \right)^2 + \left(\frac{c}{a} - \frac{b^2}{4a^2} \right) \right]$.

2. Linear over Quadratic Forms:

$$\int \frac{px + q}{ax^2 + bx + c} dx \quad \text{or} \quad \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx$$

Method: Express the numerator as: $px + q = A \cdot \frac{d}{dx}(ax^2 + bx + c) + B = A(2ax + b) + B$.

5 Integration by Partial Fractions

S.No.	Form of the Rational Function	Decomposition Form
1.	$\frac{px+q}{(x-a)(x-b)} \quad (a \neq b)$	$\frac{A}{x-a} + \frac{B}{x-b}$
2.	$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2}$
3.	$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
4.	$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
5.	$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$

6 Integration by Parts & Special Integrals

1. Standard Integration by Parts Formula:

$$\int f_1(x)f_2(x) dx = f_1(x) \int f_2(x) dx - \int \left[\frac{d}{dx}f_1(x) \cdot \left(\int f_2(x) dx \right) \right] dx$$

Priority order for selecting First Function (f_1): **ILATE** (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential).

2. Exponential Derivative Integral Identity:

$$\int e^x[f(x) + f'(x)] dx = e^x f(x) + C$$

3. Special Square-Root Integrals:

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$$

7 Definite Integrals & Fundamental Theorems

1. Fundamental Theorems of Calculus:

- **First Fundamental Theorem (Area Function):** If $A(x) = \int_a^x f(t) dt$, then $A'(x) = f(x)$ for all $x \in [a, b]$.
- **Second Fundamental Theorem (Evaluation):** If $F(x)$ is an antiderivative of $f(x)$, then:

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

2. Important Properties of Definite Integrals (P_0 to P_7):

$$P_0 : \int_a^b f(x) dx = \int_a^b f(t) dt$$

$$P_1 : \int_a^b f(x) dx = - \int_b^a f(x) dx \quad \left(\int_a^a f(x) dx = 0 \right)$$

$$P_2 : \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$P_3 : \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$P_4 : \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$P_5 : \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$$

$$P_6 : \int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

$$P_7 : \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(-x) = f(x) \quad (\text{Even function}) \\ 0, & \text{if } f(-x) = -f(x) \quad (\text{Odd function}) \end{cases}$$